Department of Computer Science & Engineering

WEEK 1 – DSA FOUNDATIONS

(17-01-2026 to 24-01-2026)

	Theory	Coding Practice:	MCQ
Day 1	Arrays: Basics, Operations, Time Complexity	1. Reverse an array / string 2. Find missing number in an array 3. Kadane's Algorithm (Maximum Subarray Sum) 4. Rotate array by K positions	Arrays and Time Complexity
Day 2	Strings: Two Pointers, Sliding Window	1. Longest common prefix 2. String compression 3. First non-repeating character 4. Sliding window: longest substring without repeating characters	Strings
Day 3	Linked Lists: SLL, DLL, Circular	1. Reverse a linked list (iterative & recursive) 2. Detect loop (Floyd's Cycle Detection) 3. Find middle of linked list 4. Merge two sorted linked lists 5. Remove nth node from end 6. Check palindrome linked list	Linked Lists
Day 4	Stacks & Queues, Applications	1. Implement stack using array / linked list 2. Stack using two queues 3. Valid parentheses 4. Next greater element 5. Infix to postfix conversion 6. Evaluate postfix expression	Stacks & Queues
Day 5	Trees: Binary Tree, BST, Traversals	1. Inorder, Preorder, Postorder traversal 2. Level order traversal 3. Height of binary tree 4. Check if tree is balanced 5. Lowest Common Ancestor (LCA) 6. Diameter of binary tree 7. BST insertion & deletion	**Binary Tree**

Day 6	Heaps & Hashing	1. Find K largest / smallest elements 2. Frequency of elements using HashMap 3. Top K frequent elements 4. Two Sum / Three Sum 5. Longest consecutive sequence	Heaps & Hashing
Day 7	Weekly Test & Assessment	• Coding execution + MCQs • Mock interview	

Recommended Daily Routine:

Students are expected to dedicate **1 hour to theory preparation**, **1 hour for coding practice**, and **1 hour for MCQs and revision** daily.

Students are mandatorily advised to use:

- **LeetCode** for coding practice
- **HackerRank** to practice MCQs and technical tests
- **GitHub account**: To maintain all the codes and projects

Placement Coordinators **Signature of HOD**